



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 • STANDARD 6.G.1

# Area of Triangles

Flagship

Lesson 5-3-flagship • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## SKY HARBOR MISSION

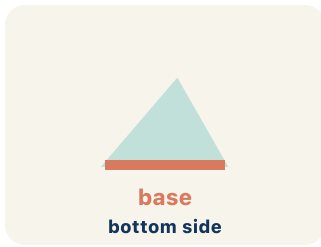
### The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



*If the bottom of a triangle is 10 cm, then  $b = 10$  cm  
in the formula  $A = \frac{1}{2} \times b \times h$*

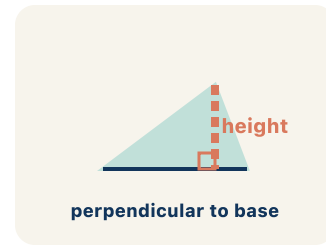
### Base

Write the definition:

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*A dashed vertical line from the top point straight  
down to the base at a  $90^\circ$  angle*

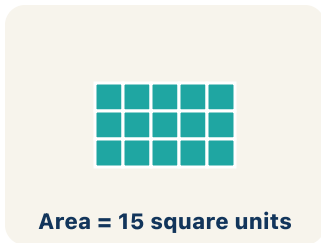
### Height

Write the definition:

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*A triangle with  $b = 8$  and  $h = 6$  has area  $= \frac{1}{2} \times 8 \times$   
 $6 = 24$  sq units*

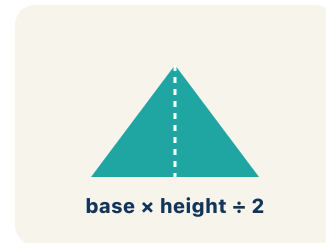
### Area

Write the definition:

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*The corner of a book or a door frame meets at  
exactly  $90^\circ$*

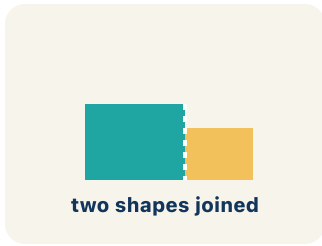
### Perpendicular

Write the definition:

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two shapes joined

*A house shape = a rectangle (walls) + a triangle (roof)*

### Composite figure

Write the definition:

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$$3x + 5$$

no equal sign

*$A = \frac{1}{2} \times b \times h$  means Area equals one-half times base times height*

### Formula

Write the definition:

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

I can find the area of a triangle using the formula  $A = \frac{1}{2} \times \text{base} \times \text{height}$ .



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Base

Height

Area

Perpendicular

Composite figure

Formula



Tap any word to see what it means and a picture.

1 The side of a triangle used as the bottom for measuring area is the

2 The perpendicular distance from the base to the opposite vertex is the

3 The amount of flat space inside a triangle is its .

4 Two lines that meet at a  $90^\circ$  angle are .

5 A shape made of two or more basic shapes combined is a

6 A math rule written with symbols, like  $A = \frac{1}{2} \times b \times h$ , is a



### Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** What is the area of a triangle with base 10 cm and height 6 cm?

- A. 30 sq cm
- B. 60 sq cm
- C. 16 sq cm
- D. 30 cm

**Step 1**  $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 6 = 30$  square centimeters.

 **Answer:** A. 30 sq cm

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

- A. 6 ft
- B. 3 ft
- C. 16 ft
- D. 12 ft

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** What is the area of a triangle with base 14 ft and height 6 ft?

- A. 42 sq ft
- B. 84 sq ft
- C. 20 sq ft
- D. 42 ft

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. A triangular flag has a base of 12 in along the pole. Its two slanted sides measure 10 in, but a dashed line shows the perpendicular height is 7 in. What is the area of the flag?**

- A. 42 sq in
- B. 60 sq in
- C. 70 sq in
- D. 120 sq in

Show your work:

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**2. Your studio designs a triangular road sign with a base of 6 ft and a height of 9 ft. What is its area?**

- A. 27 sq ft
- B. 54 sq ft
- C. 15 sq ft
- D. 27 ft

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**An architect designs a triangular roof gable with a base of 18 ft and a height of 9 ft. A second gable has the same area but a base of 27 ft. Find the area of the first gable, then determine the height of the second gable. Show all your work.**

*Sentence starter: The first gable's area is \_\_\_\_ sq ft because  $A = \frac{1}{2} \times \text{____} \times \text{____}$ . The second gable has the same area, so  $\text{____} = \frac{1}{2} \times 27 \times h$ , which means its height is \_\_\_\_ ft.*

Show your work:

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## Reflect — Exit Ticket

**A triangle has a base of 11 inches and a height of 8 inches. What is its area?**

- A. 44 sq in
- B. 88 sq in
- C. 19 sq in
- D. 44 in

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Base
2. **Notes 2:** Height
3. **Notes 3:** Area
4. **Notes 4:** Perpendicular
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **We Try (notes):** A.  $6 \text{ ft} - 24 = \frac{1}{2} \times 8 \times h \rightarrow 24 = 4h \rightarrow h = 6 \text{ ft}.$
8. **You Try (notes):** A.  $42 \text{ sq ft} - A = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42 \text{ sq ft}.$
9. **Try It 1:** A.  $42 \text{ sq in} - \text{Use the perpendicular height (7 in), not a slanted side. } A = \frac{1}{2} \times 12 \times 7 = \frac{1}{2} \times 84 = 42 \text{ square inches}.$
10. **Try It 2:** A.  $27 \text{ sq ft} - A = \frac{1}{2} \times 6 \times 9 = \frac{1}{2} \times 54 = 27 \text{ square feet}.$
11. **Exit Ticket:** A.  $44 \text{ sq in} - A = \frac{1}{2} \times 11 \times 8 = \frac{1}{2} \times 88 = 44 \text{ square inches}.$